

College of Industrial Technology

King Mongkut's University of Technology North Bangkok

Midterm Examination of Semester 2

Subject: 392132 Physics II

Date: 23 January 2020

Year: 2019

Section: 15 - 18

Time: 10.00-12.00

Seat No.

Name: _____ ID: _____ Class: _____

Directions: The test is designed to measure your comprehension. There will be 7 pages (including this page), 14 questions and they are worth 50 points.

- Answer the questions on this test paper.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators can't be used in this test.

Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

1. A body projected downward at an angle of 37° with the horizontal from the top of a building 200 m high. Its initial speed is 50 m/s.

a) What is the vertical velocity before strike the ground? (2 points)

b) How long will it take before striking the ground? (2 points)

c) How far from the foot of the building will it strike? (1 point)

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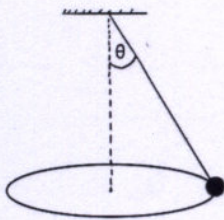
2. A baseball is thrown with an initial velocity of 100 m/s at an angle of 30° above the horizontal. How far from the throwing point will the baseball attain its original level?

(3 points)

3. An artillery shell is fired with an initial velocity of 300 m/s at 53.0° above the horizontal. To clear an avalanche, it explodes on a mountainside 40.0 s after firing. What are the x- and y-coordinates of the shell where it explodes, relative to its firing point? (3 points)

4. As show in figure, a ball is fastened to one end of a 80 cm string, and the other end is held fixed at point Q. The ball whirled in the horizontal circle shown. Find the speed of the ball in its circular path if the string makes an angle of 30° to the vertical.

(4 points)



5. A wheel rotates with a constant angular acceleration of 4 rad/s^2 . If the angular speed of the wheel is 2.00 rad/s at $t = 0$,

a) through what angle does the wheel rotate between $t = 0$ and $t = 3.00 \text{ s}$? Give your answer in radians and in revolutions. (3 points)

b) What is the angular speed of the wheel at $t = 3.00 \text{ s}$? (1 point)

6. A box rests at a point 0.5 m from the axis of a horizontal circular platform. The coefficient of static friction between box and platform is 0.2 . As the rate of rotation of the platform is slowly increased from zero, at what angular speed will the box first slide? (3 points)

7. a) What is the kinetic energy of a ball having a mass of 150 g that is thrown with a velocity of 72 km/hr ? (1 point)

b) If the baseball above was initially at rest, how much work was done on it to give this kinetic energy? (2 points)

8. How difference between gravitational energy and work by gravitational force?

(2 points)

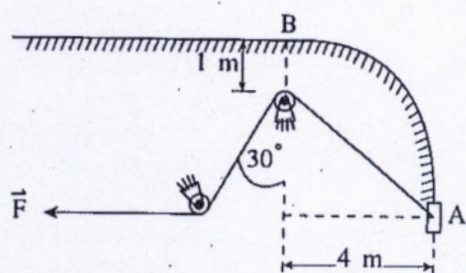
9. A box has a mass of 6 kg. The box is lifted from the garage floor and placed on a shelf. If the box gains 180 J of Potential Energy, how high is the shelf? (2 points)

10. A 60-N force is applied to a 100-kg cart at rest for 5 seconds on a frictionless surface. (on the horizontal surface)

- a) How much work has been done to the cart? (2 points)
- b) What's the average power in the first 10 seconds? (2 points)
- c) What's the kinetic energy of the cart after 20 seconds? (2 points)

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11. Force $F = 120 \text{ N}$ Pull a mass 2 kg from rest show in figure in below. What is the speed of mass when pass point B? If curve AB is circle curve and on this curve frictionless. (3 points)



12. A 2 kg book has been dropped the top of the football stadium. Its speed is 4 m/s when it is 3 m above the ground.

- a) What is its total mechanical energy? (2 points)
- b) What was the total mechanical energy of the book at the instant it was released? (1 point.)
- c) How high is the stadium? (1 point.)

13. A spring with a force constant of 500 N/m has a relaxed length of 2.5 m. When a mass is attached to the end of the spring and allowed to come to rest, the vertical length of the spring is 3 m.

a) Calculate the elastic potential energy stored in the spring. (1 point)

b) If pull down spring to length 3.5 m and release a mass, what is minimum length of spring? (3 points)

14. A boy on a bicycle drags a wagon full of newspapers constant speed at 0.80 m/s for 30.0 min using a force of 40.0 N. How much work has the boy done? and How much the average power? (4 points)

Good Luck

Asst. Prof. Dr. Prakasit Changsuphan

Asst. Prof. Dr. Dusit Ngamrunroj